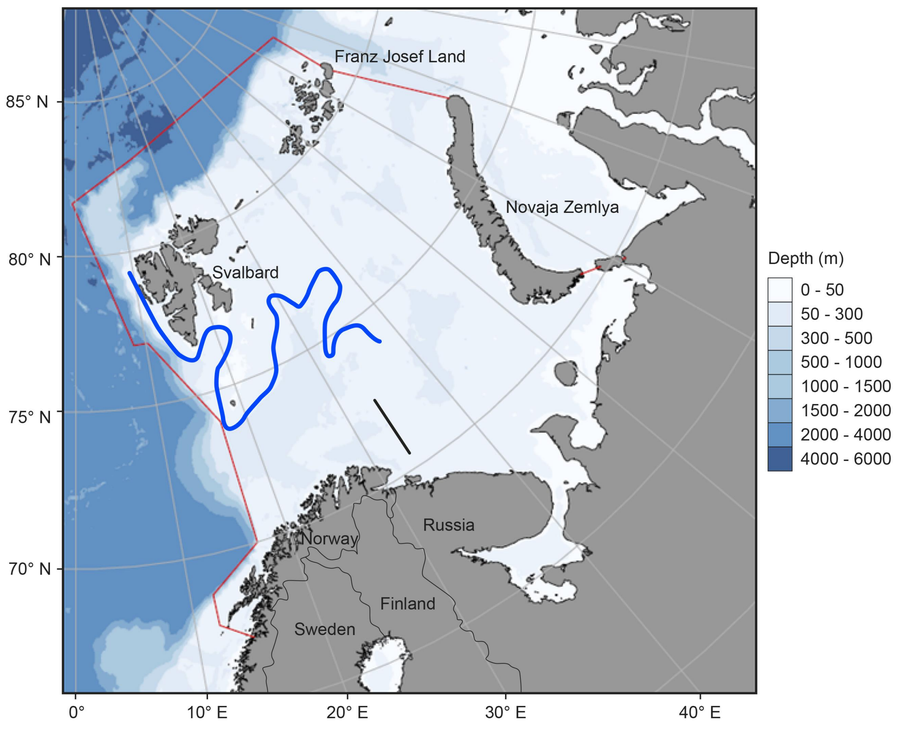
MODELLED SPECIES DISTRIBUTIONS AND CHANGES IN WATER TEMPERATURE AND ICE COVERAGE IN THE BARENTS SEA IN THE LAST DECADE

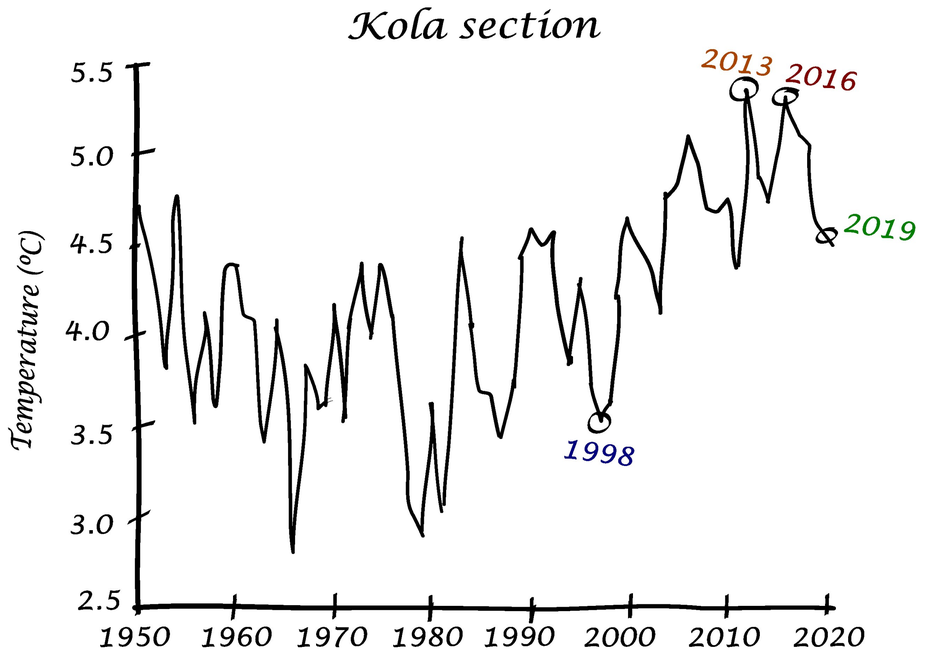
Nascimento M.C.[1], Husson B.[2], Guillet L[2]. and Pedersen, T[1]

[1]Department of Arctic and Marine Biology, UiT The Arctic University of Norway, Tromsø, Norway

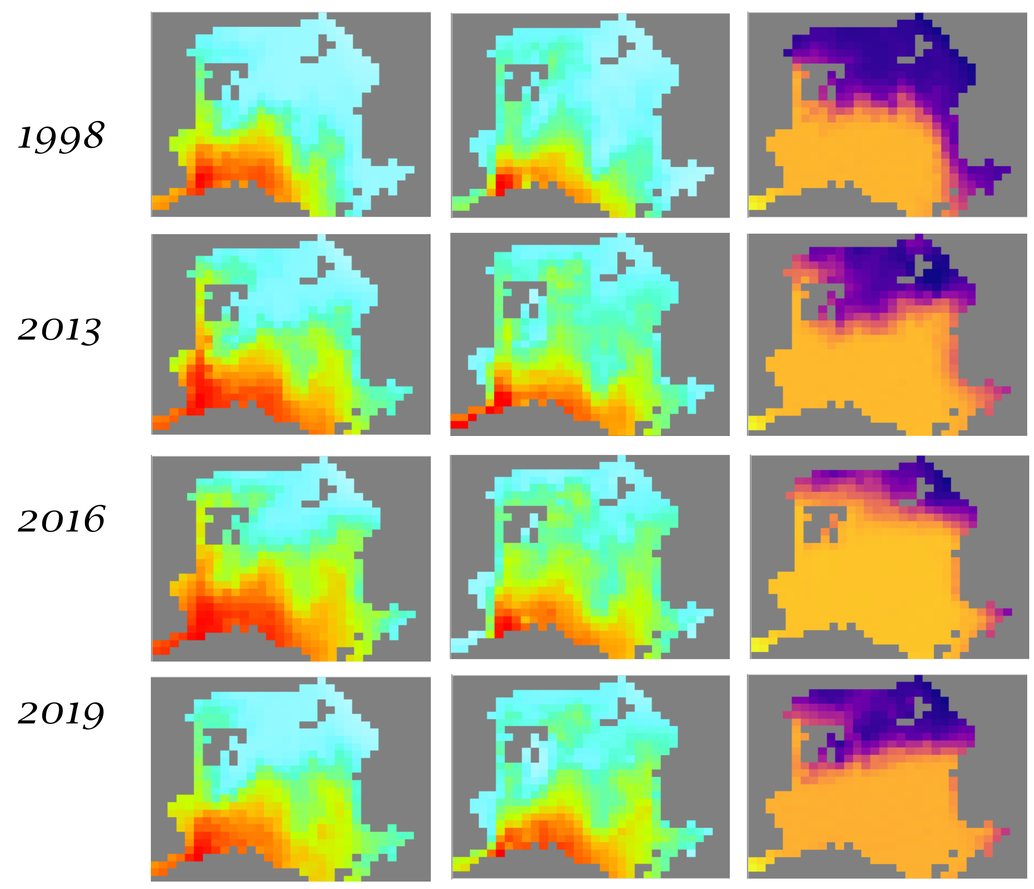
[2]Norwegian Institute of Marine Research, Bergen, Norway

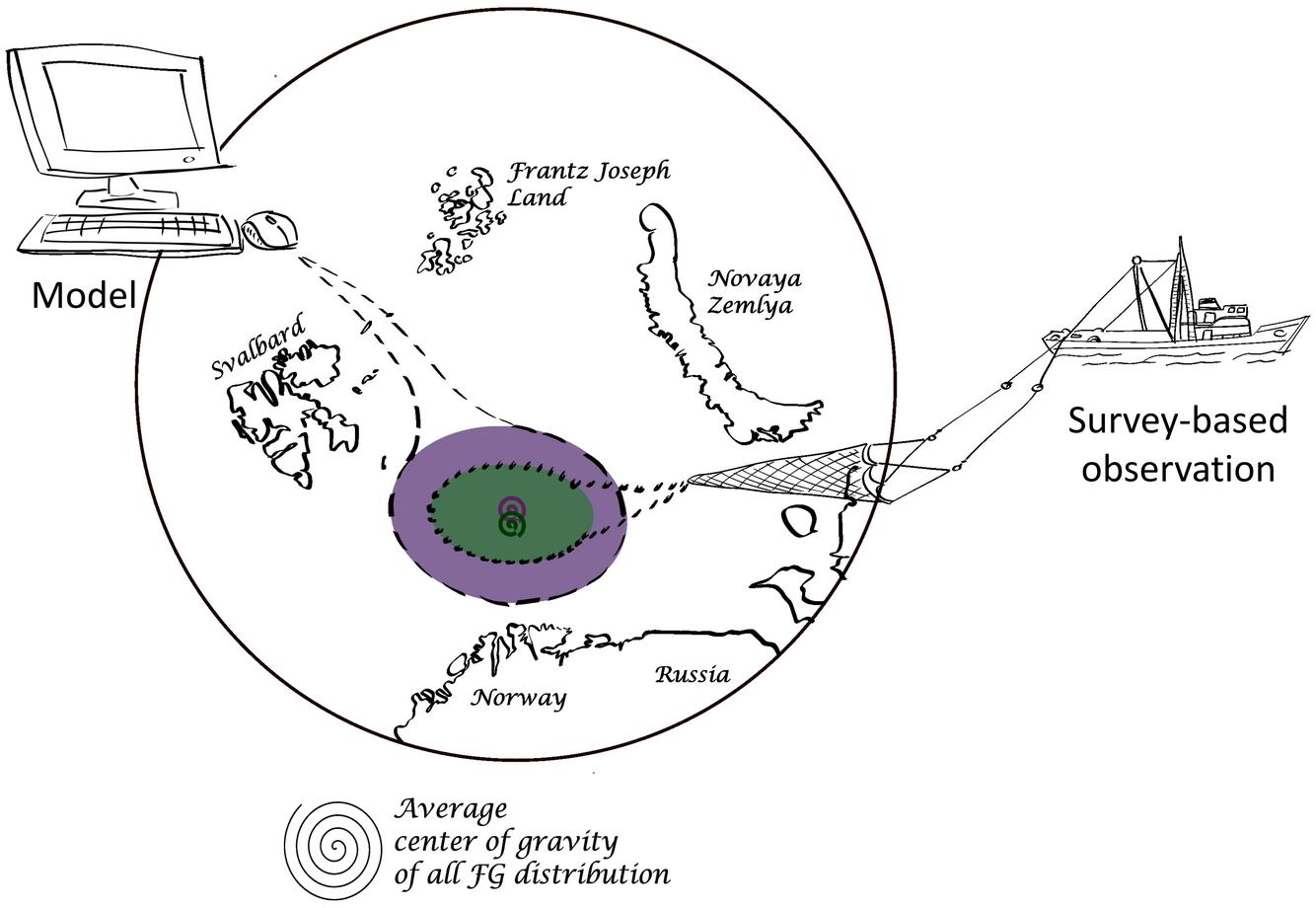


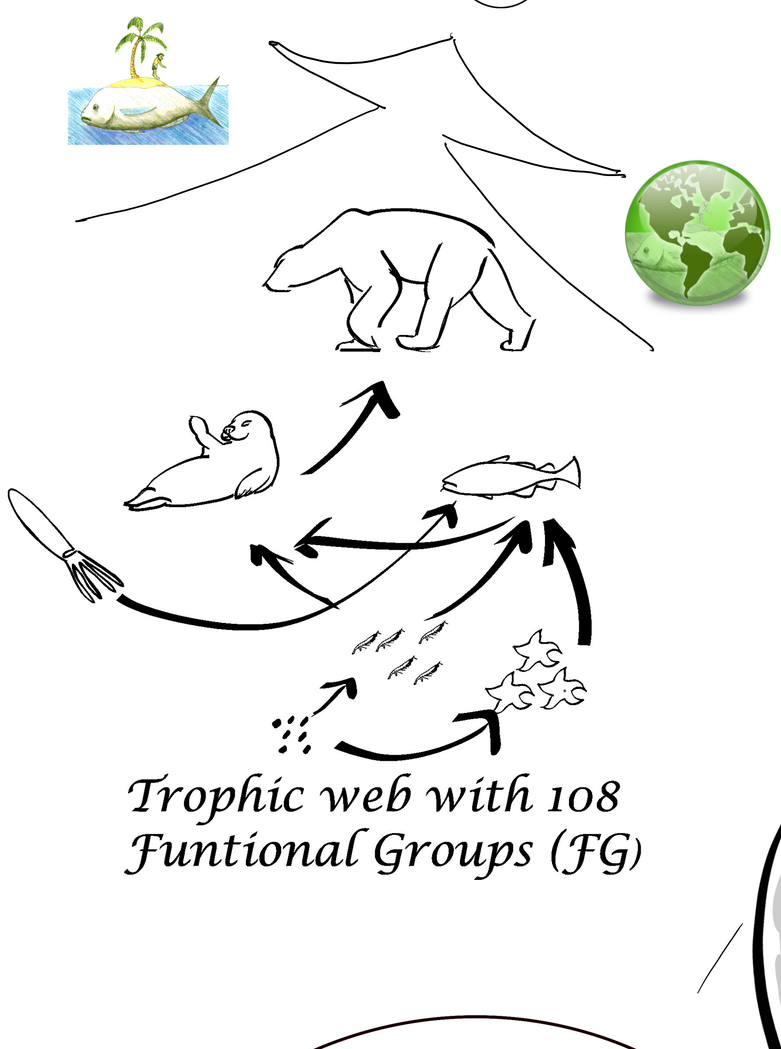
In the Barents Sea, the last decade (based on Kola section 0-200 m measurements) was 0,26ºC warmer than the previous decade (2000-2009) and 0,72 ºC warmer than the decade from 1990 to 1999.



We built an Ecopath with Ecosim (EwE) model with 108 functional groups (FG) representing the Barents Sea (BS) ecosystem (Pedersen et al., 2021). Then we created a spatial model using the Ecospace plugin of the EwE software. We adjusted our model with environmental drivers for cold and warm years .

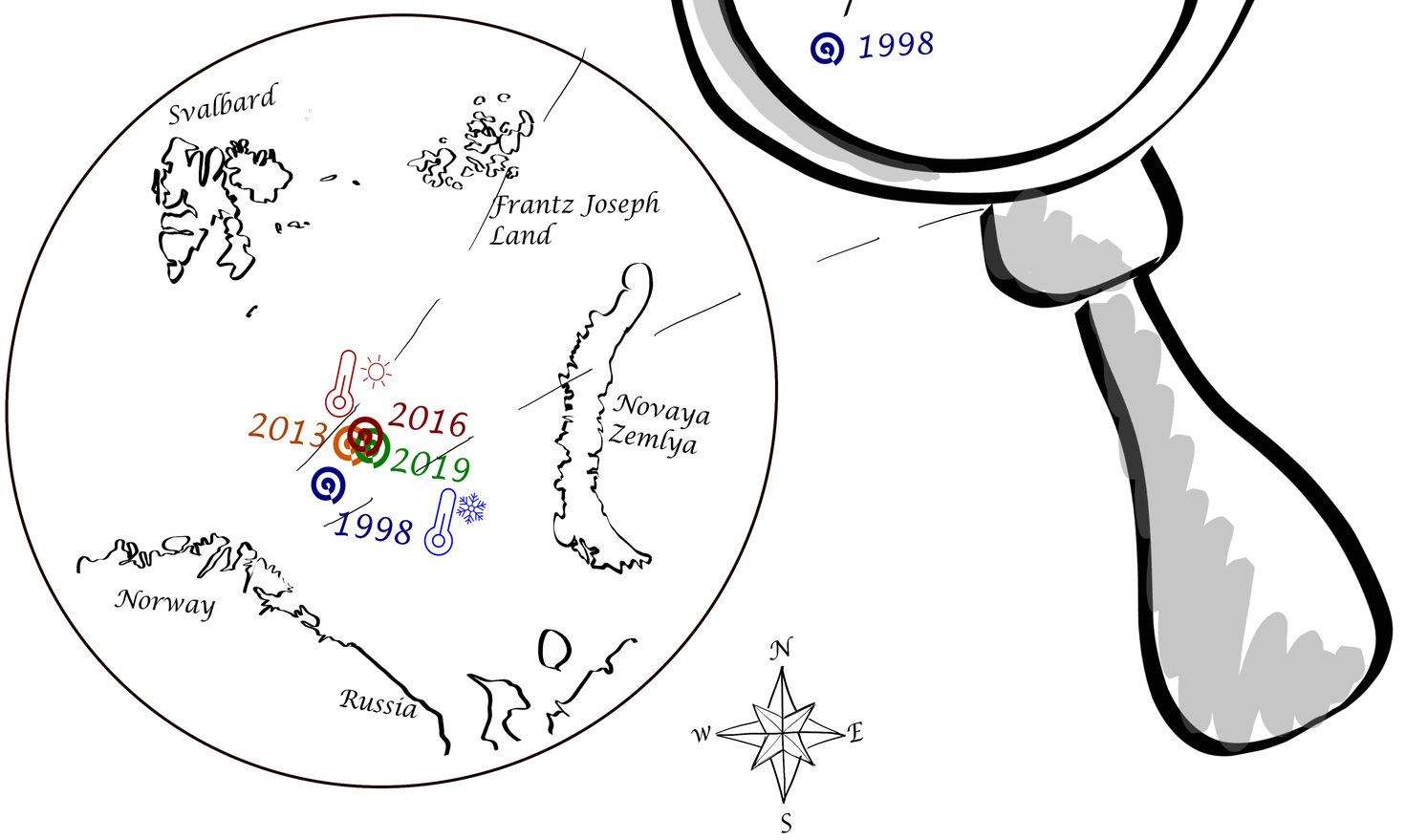






Averagely we observed a high correspondence between the modeled and observed distributions when comparing the predicted and natural distribution's center of gravity and inertia.

We projected the distributions of the same FGs for 2018, 2013, 2016, and 2019.



On average, the functional groups are shifting the distributions north and northeastern, showing that with the increase in temperatures and reduction in the ice coverage in the Barents Sea, many species are moving poleward.